

AI POWERED INTELLIGENT SALES FORECASTING

TEAM MEMBERS:

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Why AI in Sales Forecasting?

- Sales data is growing rapidly and is difficult to analyze manually.
- Traditional forecasting may fail to capture complex sales patterns.
- AI identifies trends and relationships hidden within historical data.
- Machine Learning helps predict future sales more accurately.
- AI enables faster, data-driven business decisions.
- Forecasting supports better planning of sales, inventory and resources.

Predicting Future Sales with Artificial Intelligence

Tech Roles

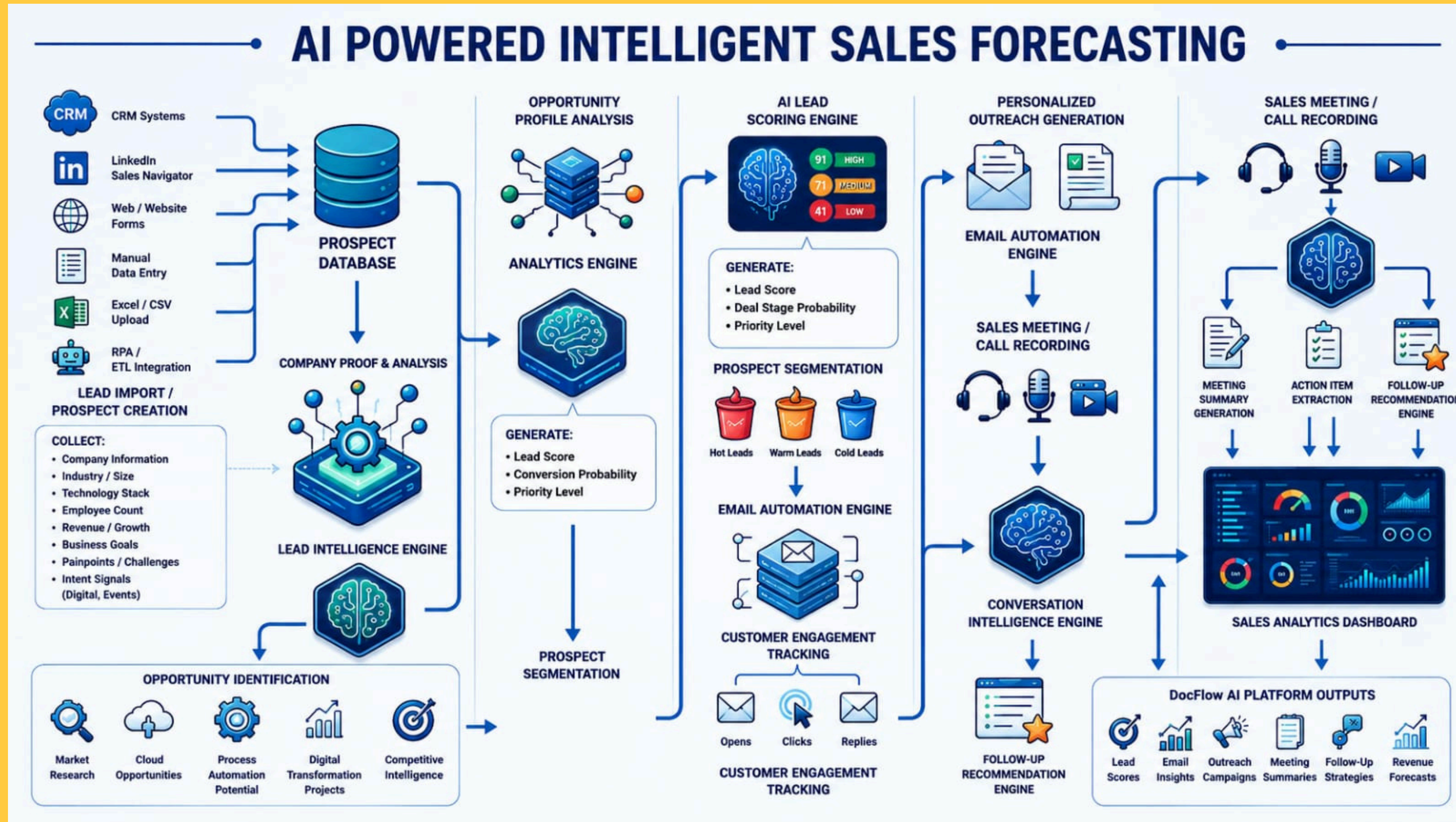
1. Frontend Developer
2. Backend Developer
3. AI/ML & LLM Developer
4. Database & CRM Integration Developer
5. Testing, DevOps & Documentation

Different skills, one team, one goal.

TECHNOLOGIES & TOOLS USED

Tech Role	Tools Used
Team Lead	Git, GitHub
Backend Developer	Python, FastAPI, SQLAlchemy
Frontend Developer	Python, Streamlit
AI / ML Engineer	Pandas, NumPy, Scikit-learn, Google Gemini
Database Developer	PostgreSQL, SQLAlchemy
QA & Testing	Swagger UI, API Testing
DevOps	Docker, Docker Compose

SYSTEM ARCHITECTURE



Modules Overview

MODULE 1

- Build lead import (CSV upload, manual entry) with duplicate detection
- Build Streamlit 'Leads' tab: prospect search list + detail panel

MODULE 2

- Implement company profile analysis logic (industry, size, tech stack ingestion)
- Build AI qualification scoring prompt + Lead Intelligence card generation
- Integrate Lead Intelligence output into Leads UI

MODULE 3

- Build AI email/outreach generation service (prompt templates per industry)
- Build Streamlit 'Outreach' tab: AI email Generator UI with edit-before-send

MODULE 4

- Implement lead scoring & conversion probability model(rules/ML features)
- Build recommendation engine: follow-up timing, channel mix, content strategy
- Integrate Lead Score + Outreach Strategy cards into Outreach UI

MODULE 5

- Build FastAPI endpoints for CRM integration (SalesForce/Hubspot schema mapping)
- Implement CRM sync logic (contacts,, activites, deal stages) + sync logs
- Build meeting/call summarization (key points + action item extraction) via LLM
- Build Stramlit ‘Conversations’ tab: CRM Sync status + Meeting Summary + Activity feed

MODULE 6

- Build Sales Analytics Dashboard (conversion rate, pipeline value, sales cycle KPIs)
- Implement Kanban-style pipeline view + automated follow-up recommendations panel

Module 1: Lead Management

Lead Import & Management

Built a streamlined lead import system supporting CSV uploads and manual lead entry, with duplicate detection to maintain clean and reliable lead data.

Leads Tab

Developed a Streamlit-based Leads interface with prospect search, an organized lead list, and a detailed lead information panel for quick access and management.

Screenshots



SG

SalesGenie AI

Sales assistant & lead intelligence

Leads

Add Lead

Company Intelligence

Outreach

Lead Scoring

Conversations

Dashboard

P

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Account

Log out

Deploy

Add Lead

Import leads in bulk from a CSV file, or add a single lead manually.

CSV ImportManual Import

IMPORT LEADS FROM CSV

Upload a CSV file containing your lead information. Duplicate leads based on email or phone will be skipped.

Choose CSV file

Upload

200MB per file • CSV

samples_leads.csv

467.0B

+

Deploy

Found 4 lead(s) in the CSV file.

Import CSV Leads

Import complete.

TOTAL ROWS

4

CREATED

3

DUPLICATES SKIPPED

1

ERRORS

0

ROW	STATUS	COMPANY	DETAIL
2	DUPLICATE	NovaTech Solutions	A matching lead already exists.
3	CREATED	BrightWave Systems	—
4	CREATED	CloudNest Technologies	—
5	CREATED	Vertex Digital Solutions	—

Refresh the Leads list to see the imported leads.

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Leads

Tech SolutionsNEWView

MediCare SolutionsNEWViewing

BlueSky RetailNEWView

vision vaultNEWView

GreenEdge RetailQUALIFIEDView

MediCare SolutionsNEW

BASIC LEAD INFORMATION

Company nameLocation

MediCare Solutionsrtpp

Contact nameIndustry

sowmyaHealthcare

EmailCompany size

sowmi@gmail.com201-500

PhoneJob title

1234567894IT Manager

Lead Records Directory

A complete record of every lead in the system, with full profile details.

GreenEdge RetailQUALIFIEDView

Lead Records Directory

A complete record of every lead in the system, with full profile details.

ID	COMPANY	CONTACT	EMAIL	PHONE	INDUSTRY	LOCATION	COMPANY SIZE	LEAD SOURCE	BUDGET	S
4	string	string	user@example.com	string	string	string	string	string	string string	
1	string	string	user1@example.com	string	string	string	string	string	string string	
7	Tech Solutions	Arun Kumar	arunkumar23@test.com	9876543210	Software	Chennai	100	Website	USD 20000	
8	MediCare Solutions	sowmya	sowmi@gmail.com	1234567894	Healthcare	rtp	201-500	Website	INR 1000000	
10	BlueSky Retail	Wanda	wanda@testexample.com	9876543213	Retail	Chennai	51-200	Website	INR 1500000	
5	vision vault	perumal	ramesh@gamil.com	1234567891	Hospitality	chennai	200-500	website	INR 1000000	
12	GreenEdge Retail	Vikram Nair	Vikramnair@gmail.com	5465498469	Retail	kochi	201-500	LinkedIn	INR 1200000	
13	BrightWave Systems	Karthik Menon	karthik.menon@example.com	9123456780	Software	Chennai	—	—	—	
14	CloudNest Technologies	Meera Shah	meera.shah@example.com	9988776655	Cloud Services	Hyderabad	—	—	—	

MODULE 2 — COMPANY & LEAD INTELLIGENCE

Company Profile Analysis

Built the company profile analysis logic to ingest and process key company information such as industry, company size, and technology stack.

AI Qualification & Lead Scoring

Developed the AI qualification scoring logic to evaluate leads and generate meaningful lead intelligence based on the available company and lead information.

Lead Intelligence UI Integration

Integrated the generated intelligence directly into the Leads interface through a score badge and insights panel, making AI-driven information easy for users to view and understand.

Screenshots



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Lead Intelligence & Company Analysis

Analyze company profiles, identify business needs and opportunities, and generate lead qualification scores.

Analyze Company

Uses the same AI lead-scoring model as Outreach → Score Lead, framed as company-level insight.

Select a lead to analyze

BlueSky Retail (Wanda)

Run Analysis

Uses the same AI lead-scoring model as Outreach → Score Lead, framed as company-level insight.

Select a lead to analyze

BlueSky Retail (Wanda)

Run Analysis

LEAD INTELLIGENCE

Overall Lead Score

85

OUT OF 100

Highly Qualified

Company Snapshot

Company

BlueSky Retail

Industry

Retail

Engagement

High

Conversion Probability

85%

LOW MEDIUM HIGH

High Probability

AI Insight

BlueSky Retail is a highly qualified inbound lead with a substantial budget (\$1.5M) and a clear operational pain point in manual customer management that aligns strongly with automated CRM/sales solutions.

Recommended Opportunity

Schedule a product demo focused on automated customer management and CRM capabilities.

MODULE 3 – AI EMAIL OUTREACH

AI Email Generation

Built an AI-powered email outreach service to generate personalized sales emails based on lead information.

Industry-Specific Prompts

Created prompt templates tailored to different industries to make the generated outreach more relevant and targeted.

Outreach Interface

Developed a streamlined Outreach tab with an AI email generator, allowing users to review, edit the generated email before sending.

Screenshots



SGSalesGenie AISales assistant & lead intelligence

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AI Outreach Generation

Generate personalized outreach emails and follow-ups that convert prospects into real meetings.

Select Lead

Choose a lead to generate outreach.

WABlueSky Retail (Wanda)

View Lead Details

Personalization Inputs

Add details to personalize your email.

Tone of Email

Professional

Email Length

Medium

Email Type

Cold email

Primary Goal

Book a Demo

Key Pain Points

Manual customer management X

+ Add

Business Goals

Increase sales X

+ Add

Generate Mail

Generate your personalized outreach email.

Generate Email

AI Generated Email

Refresh

To wanda@testexample.com

Subject: Unlocking omnichannel growth and boosting sales for BlueSky Retail

Hi Wanda,

Thanks for visiting our website recently. As Sales Manager at BlueSky Retail, driving revenue while delivering exceptional customer experiences across every channel is likely top of mind.

Many retail leaders find that manual customer management creates operational bottlenecks, pulling valuable time away from inventory optimization and scaling omnichannel growth. That's where SalesGenie comes in.

Our AI-powered platform seamlessly integrates with your current Salesforce CRM to automate routine customer management tasks. By streamlining lead follow-ups and surfacing actionable buyer insights, SalesGenie frees your team to focus on high-impact selling, elevating the shopper experience and driving higher

Copy to Clipboard

Edit Email

Send Email

MODULE 4 — LEAD SCORING & RECOMMENDATIONS

Lead Scoring & Conversion Prediction

Implemented an ML-based lead scoring and conversion probability model using relevant lead features to identify high-potential prospects.

Recommendation Engine

Built a recommendation engine that suggests the most suitable follow-up timing, communication channel mix, and content strategy for each lead.

Outreach Strategy Integration

Integrated the lead score and recommended outreach strategy into the Outreach UI through easy-to-understand strategy cards.

Screenshots



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Deploy

Select Lead

Choose a lead to analyze and score

BlueSky Retail (Wanda)

Analyze Lead

Click Analyze Lead above to generate the AI score, recommendation, and best actions.

Deploy

Score

AI predicted conversion likelihood

80

OUT OF 100

Highly Qualified

Engagement Score

80 / 100

Fit Score

72 / 100

Intent Score

78 / 100

Decision Readiness

80 / 100

Recommendation

What our AI suggests based on lead analysis

BlueSky Retail is a highly qualified inbound lead with a clear budget and an urgent 3-month timeline to resolve manual customer management issues. Engage Wanda promptly to understand their specific workflow needs and identify key economic decision-makers.

MODULE 5 – CRM INTEGRATION & AI CONVERSATIONS

CRM Integration

Built FastAPI endpoints for CRM integration, including Salesforce and HubSpot schema mapping to connect external CRM data with the application.

CRM Synchronization

Developed the AI qualification scoring logic to evaluate leads and generate meaningful lead intelligence based on the available company and lead information.

Lead Intelligence UI Integration

Integrated the generated intelligence directly into the Leads interface through a score badge and insights panel, making AI-driven information easy for users to view and understand.

MODULE 6 — SALES ANALYTICS & PIPELINE

Sales Analytics Dashboard

Built a sales analytics dashboard to track key KPIs including conversion rate, pipeline value, and sales cycle performance.

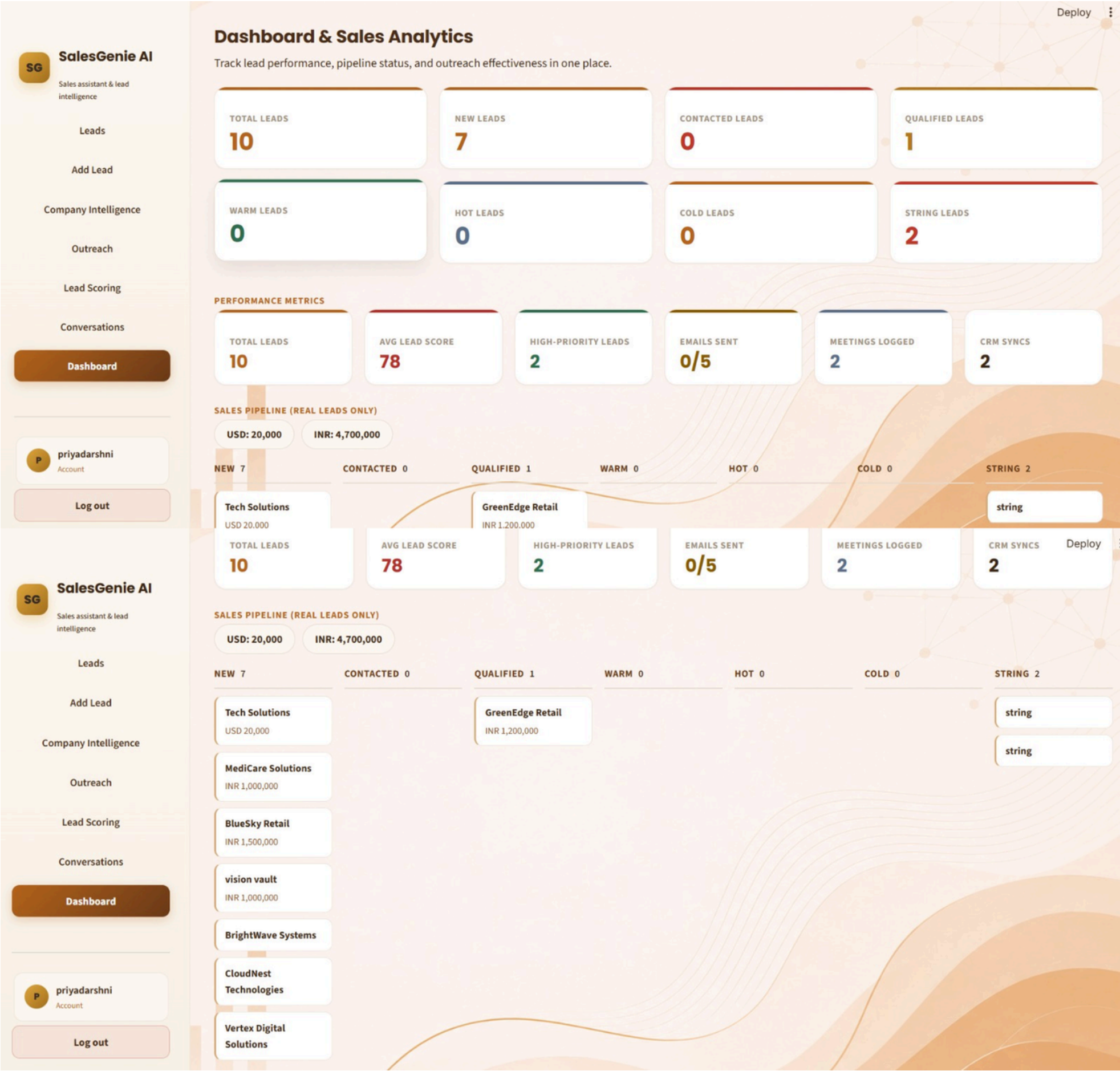
Kanban Pipeline View

Implemented a Kanban-style sales pipeline to visually track leads and deals across different stages.

Automated Follow-Up

Added automated follow-up capabilities with integration panels to help sales teams manage follow-ups efficiently.

Screenshots



SYSTEM INTEGRATION

Objective: Integrate the individual modules into a unified AI-powered sales forecasting platform.

Key Integration Areas:

- Integrated frontend with backend APIs
- Connected lead management with AI intelligence
- Integrated AI-based lead scoring and conversion prediction
- Connected AI email generation with lead information
- Integrated CRM synchronization through FastAPI
- Connected database, AI services and user interface
- Validated end-to-end data flow between modules

Integration Flow

User → Streamlit UI → FastAPI → Database / AI Services → Response → UI

End-to-End Integration Workflow

Lead Entry / CSV Upload



Lead Management



Company Analysis



Lead Intelligence



Lead Scoring



AI Email Generation



CRM Integration



Sales Analytics

All modules communicate through the integrated frontend, backend APIs, database and AI services to provide a seamless sales intelligence workflow.

TESTING & QUALITY ASSURANCE

- **Module Testing** — Verified each module's core functionality.
- **UI Testing** — Checked Streamlit tabs, forms, buttons, search and panels.
- **API Testing** — Tested FastAPI endpoints using Swagger UI.
- **Integration Testing** — Verified communication between frontend, backend and database.
- **Data Validation** — Checked lead inputs, duplicate detection and generated outputs.
- **AI Feature Testing** — Verified lead scoring, recommendations and AI-generated emails/summaries.
- **Error Handling** — Identified failures and verified fixes through re-testing.

THANK YOU

“Predict the future. Understand the customer. Act smarter.”